

Frictions & the MM benchmark

Week 36 — corporate finance as value minus frictions

Advanced Corporate Finance (EBM232A05)

University of Groningen · Sebastian Pfeil

Today

- What (advanced) corporate finance is — and whose objective it serves.
- The organising idea of the whole course: $\text{value} = \text{benchmark} - \text{frictions}$.
- The benchmark (Modigliani–Miller) — fast, you already own it.
- Two puzzles that show how little we actually explain.
- The first twist: even **taxes** are subtler than you were taught.
- How the term maps onto a cumulative **frictions ledger** — and what is new vs. what you already know.
- How this course works.

A theory-forward course: every week, a friction, a small model with an insight, the evidence.

What is corporate finance?

The corporation — five characteristics

A corporation is a specific legal technology (Hansmann and Kraakman, 2004):

1. **Legal personality** & permanence — it outlives its owners.
2. **Delegated management** — a board and managers, not the owners, decide.
3. **Share ownership** by investors — who can diversify.
4. **Transferable** shares — ownership trades without disrupting the firm.
5. **Limited liability** for investors.

A vehicle for *large-scale, risky* operations: diversified owners, professional managers, tradable claims. Corporate finance studies the **financial** side of that vehicle.

This is an old problem (a little history)

Bazacle, Toulouse (1372). A merger of French milling companies with outside shareholders, annual meetings, tradable shares, and **dividends** — six centuries ago (Le Bris et al., 2019).

VOC (1602). The Dutch East India Company: a charter to fund risky long-distance trade, soon made effectively **permanent**, with a clean **separation of shareholders and management** (Gelderblom et al., 2013).

The institution is old; the *financial* questions — how to fund it, who bears the risk, who decides — are the ones we still cannot fully answer.

The corporate-finance decisions

Management acts for the providers of capital across a handful of recurring decisions:

- **Investment** — capital budgeting, M&A. (*Creates value.*)
- **Financing** — the debt/equity mix and its many forms.
- **Payout** — dividends vs. repurchases.
- **Risk management** — hedging operational and financial risk.
- **Governance** — the control and incentive structure around all of it.

Investment *creates* value; the financial decisions largely *protect* it from frictions. This course is mostly about the second kind.

Whose objective?

Whose interests does the firm serve?

Shareholder value. Shareholders are the **residual** claimants — everyone else is paid first — so maximising their value gives a *single, well-defined* objective (the Friedman 1970 view).

Stakeholders. Most corporate laws require boards to **balance** stakeholders; Graham (2022) finds CFOs shifting away from pure shareholder focus, toward stakeholders such as employees and customers; Mehrotra and Morck (2017) weigh the two.

Our working objective: maximise firm value. It is tractable, and — as we will see — most stakeholder concerns re-enter as *frictions* on that objective, not as a rival goal.

And where does ESG fit?

The 2020s version of the same question — read it with this course's tools:

- Most value-relevant sustainability is **just NPV**: a long-horizon investment paying off late — Edmans's **rational sustainability**. No rival objective needed.
- The genuine exception: the **externality** — a cost the firm imposes but does not bear. It never re-enters through NPV; **regulation** prices it, or nobody does. Week 42 works one out in full: the **taxpayer** as the bank's involuntary stakeholder.
- So the question for any ESG claim is never “is it good?” but “**which friction, and out of whose pocket?**”

Reading: Edmans (2024) (short). Stakeholder *governance* — boards, codes, voice — is EBM811C05's territory.

The organising idea: value minus frictions

The one question we ask all term

Which market friction pushes real firms off the frictionless benchmark — and in which direction?

We make this precise with a single accounting identity for firm value that we extend, one friction at a time. Start from the frictionless world.

Perfect world: value from real decisions only

With perfect markets, firm value is just discounted cash flow:

$$V = \frac{X}{K}, \quad X = \text{expected cash flow}, \quad K = \text{cost of capital}.$$

- **All** value comes from *real* (investment) decisions.
- **Financial** decisions — the debt/equity mix, payout, hedging — add *nothing* (Modigliani and Miller, 1958).

The null. Everything interesting is a *deviation*. (In the unlevered firm $K = r_A$ — the same r_A we use in the MM recap, so $V_U = X/r_A$.)

Subtract friction 1: inefficiency costs

Relax “perfect markets”. Financial decisions now move value by changing **deadweight costs**:

$$V = \frac{X}{K} - \underbrace{\text{INEFFICIENCY COSTS}}_{\text{taxes} \cdot \text{distress} \cdot \text{mispricing}}$$

- Taxes: interest is deductible — a **benefit** of debt.
- Financial distress: bankruptcy is **costly** (10–20% of value) — a cost of debt.
- Mispricing: securities can be issued cheap or dear — timing.

Financial policy earns its keep by *minimising* these costs. (Weeks 37–38, 43.)

Subtract friction 2: agency costs

Relax “managers maximise shareholder value / complete contracts”. Interests diverge (Jensen and Meckling, 1976):

$$V = \frac{X}{K} - \text{INEFFICIENCY} - \underbrace{\text{AGENCY COSTS}}_{\text{monitoring} + \text{bonding} + \text{residual}}$$

- **Manager vs. shareholder:** empire-building, quiet life, pay.
- **Creditor vs. shareholder:** risk-shifting, debt overhang.

Financing, payout, and governance are *devices* to shrink these costs. (Weeks 39, 41, 43.)

Subtract friction 3: behavioural costs

Relax “rational decision-makers”. Managers and investors err systematically (Shefrin, 2018):

$$V = \frac{X}{K} - \text{INEFFICIENCY} - \text{AGENCY} - \underbrace{\text{BEHAVIOURAL COSTS}}_{\text{biases} \cdot \text{heuristics} \cdot \text{framing}}$$

- **Biases** (over-optimism, overconfidence), **heuristics** (anchoring), **framing** (loss aversion).
- Shows up in over-investment, ill-timed M&A, sticky payout.

A lighter thread here than agency, but real — it returns in M&A (Week 41) and payout (Week 43).

The spine of the course

$$V = \frac{X}{K} - \left(\underbrace{\text{inefficiency}}_{\text{markets}} + \underbrace{\text{agency}}_{\text{incentives}} + \underbrace{\text{behavioural}}_{\text{psychology}} \right) \text{ costs}$$

- The three cost families are the three classic **lenses** on corporate finance.
- Every week isolates **one** friction, models it, and reads the evidence.
- We keep score in a cumulative **frictions ledger**: which cost, which direction.

The recurring question — **which friction, and in which direction?** — is now literally “which term in this equation, and what sign?”

The benchmark you already own

Modigliani–Miller, in one breath

You already proved this. In a perfect capital market:

$$V_L = V_U, \quad r_E = r_A + \frac{D}{E} (r_A - r_D).$$

- Leverage just reshuffles risk — **cheap debt is exactly offset** by a higher cost of equity.
- Home-made leverage lets investors undo any financing choice for free.

Miller (1988): “showing what *doesn't* matter can also show, by implication, what does.”
The rest of the course is the “what does”.

You already know the first friction: taxes

Interest is tax-deductible; equity payments are not. With a corporate tax τ_c on permanent debt D :

$$V_L = V_U + \tau_c D, \quad \text{gain from leverage} = \tau_c D.$$

- The gain rises in D with no offset — so the model says **lever up to 100%**.
- Reality: firms are nowhere near 100% debt; many are strikingly **conservative**.

Is capital structure, then, a solved problem? There are two reasons to think not.

The puzzle: how little we explain

Reason 1 — we barely explain leverage at all

Lemmon et al. (2008) run the standard leverage regression — profitability, size, tangibility, market-to-book — and add a **firm fixed effect**.

- The fixed effect alone explains the **majority** of the variation.
- A firm's leverage today predicts its leverage **two decades** later.
- The textbook determinants add surprisingly little on top.

Most of capital structure is a **stable, firm-specific “something”** we have not named. That “something” is what the frictions map is hunting for.

Reason 2 — and you already half-know it

You saw Korteweg (2010) in your bachelor's course: the *net* benefit of leverage, measured directly, is

- **hump-shaped** — a real interior optimum, not the 100% corner;
- and the median firm sits **below** it, adjusting only **sluggishly**.

Your bachelor's course left that as a **fact**. This course asks *why* — and, it turns out, even the **sign** of the tax effect is not safe.

Two puzzles, one lesson: the “obvious” friction needs a second look. Start with taxes themselves.

**The twist: even taxes are subtler
than $\tau_c D$**

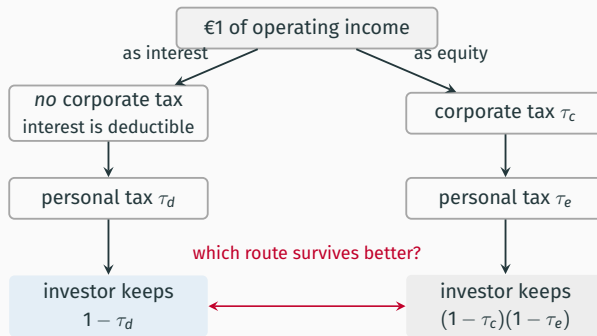
Who really pays the tax?

Your bachelor courses counted only the **corporate** tax. But investors pay **personal** taxes too, and debt and equity income are taxed **differently**.

- Interest income is taxed at the personal rate τ_d (ordinary income).
- Equity income (dividends + deferred capital gains) is taxed at an effective rate τ_e , typically **lower** than τ_d .

Miller (1977) asks: once we net out *all three* taxes, what is the true gain from a dollar of debt?

One euro, two routes



Debt skips the corporate layer but lands in the **highest** personal bracket; equity is taxed **twice**, gently the second time. The gain from leverage is the **ratio** of the two survival rates.

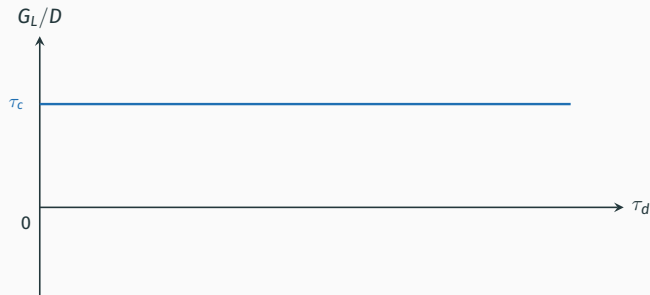
Route one dollar of income to investors: as debt, they keep $(1 - \tau_d)$; as equity, $(1 - \tau_c)(1 - \tau_e)$ — after *all* taxes. With permanent debt the interest stream ties one-for-one to D , so the gain is

$$G_L = \left[1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_d} \right] D$$

Put in numbers ($\tau_c = 25\%$, $\tau_e = 15\%$):

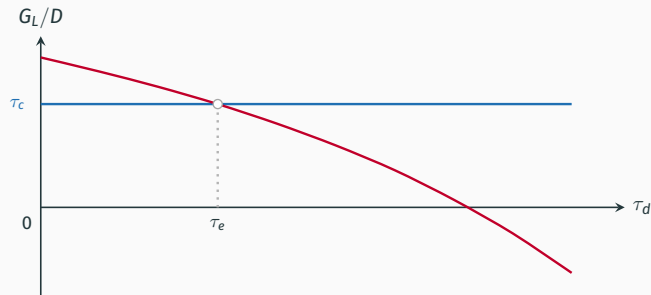
- $\tau_d = 30\%$: $G_L/D = +9\%$ $\uparrow$ — debt favoured.
- $\tau_d = 40\%$: $G_L/D = -6\%$ $\downarrow$ — debt *penalised*. The sign **flips**.
- Indifference at $\tau_d \approx 36\%$: **structure is irrelevant again**, taxes and all.

The sign flip, as a picture



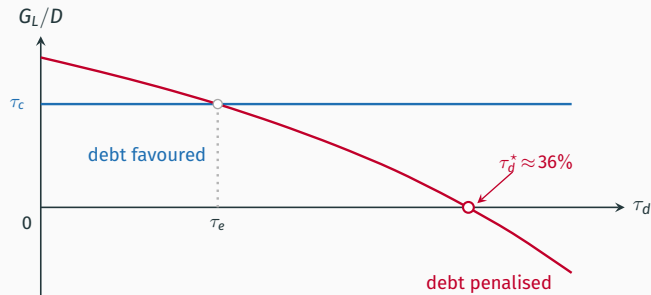
Last year's answer, drawn: the gain is τ_c per euro of debt — **flat**, and entirely indifferent to who ends up holding the debt.

The sign flip, as a picture



Net out personal taxes and the true gain **falls** as τ_d rises. It coincides with τ_c only in the knife-edge case $\tau_d = \tau_e$ — when debt and equity income are taxed alike.

The sign flip, as a picture



Past $\tau_d^* \approx 36\%$ the gain turns **negative**: the personal-tax penalty more than claws back the corporate shield. **The friction you were surest about changes sign.**

The friction you were **most** sure about — “taxes mean lever up” — can net to **zero** in equilibrium once you ask who ultimately pays the tax. Historically $\tau_d > \tau_e$, so the personal-tax penalty on debt claws much of the corporate shield back.

Two consequences we carry forward:

- “Identify the friction and sign it” is **real work**, not bookkeeping — a friction can reverse or vanish at the margin.
- If the cleanest friction is this slippery, the stable “something” in Lemmon et al. (2008) must come from **other** frictions. That is the rest of the course.

Miller’s equality is an aggregate, marginal-investor claim — tax clienteles blur it in practice. That caveat is itself a later friction.

Aside — how to read a theory paper

You will read one short theory paper most weeks. A reliable recipe:

1. **Big picture** from the introduction — what puzzle, what claim?
2. The **set-up**: players, explicit assumptions, and *what is left out* (the implicit ones).
3. The **consequences**: the solution, and above all the **driving mechanism**.
4. The **reach**: specific to the set-up, or general? How would you *frustrate* the mechanism?

Miller (1977) is a clean first target: the mechanism is one ratio of tax wedges. Try step 4 on it.

The map

The course as a map of frictions

Every week fills in one term of $V = X/K - (\text{costs})$.

Wk	Cost family	Friction	The decision it bends
36	inefficiency	personal taxes (Miller)	today's warm-up
37	inefficiency	dynamics / flexibility	how much debt, and when
38	inefficiency	costly external finance	investment; the macro link
39	agency	manager $\neq$ shareholder	incentives & pay
41	agency/beh.	market for corporate control	mergers
42	inefficiency	hedging & safety nets	risk management, regulation
43	agency/beh.	commitment & sentiment	payout

Taxes were the warm-up. The frictions that *stick* — agency, information, constraints, commitment — are ahead.

The frictions ledger — we build this all term

One row per week. By Week 43 it is your revision sheet.

Wk	Friction	Value effect	Evidence
36	Personal taxes (Miller)	↑ / ↓	net benefit (Korteweg, 2010)
37	...	—	—

The recurring question, every lecture: **which friction, and in which direction?**

What is new here — and what is not

You already own (from the bachelor):

- MM, the tax shield, the 100% corner
- trade-off, distress, pecking order
- options, real options, vega

This course adds:

- the **manager-vs-shareholder** wedge
- **dynamics** and the **macro** link
- **regulation**, and the frontier **puzzles**

The theory re-uses tools you have (option payoffs, real options) to reach results you have *not* seen. What is new is the destination, not the machinery.

How this course works

- **Seven frictions weeks** (36–43; Week 40 is Career Week — no sessions). Each: a lecture (Tue) + a tutorial (Wed).
- **Lecture** — the friction, a short derivation with an insight, the evidence.
- **Tutorial** — calculation practice on the week's model, and the project's methods thread (regression, event studies, WRDS).
- Valuation is **not** here — it lives in the separate Corporate Valuation block.

Bring the frictions ledger to every lecture. We add a row each week.

Assessment — 80% exam, 20% project

Supervised exam (80%)

Calculation-heavy: the week's **Mickey-Mouse models** as numeric drills, plus friction-identification and evidence-interpretation. Closed-book, non-programmable calculator.

Replication project (20%)

In teams of five, **reproduce the headline table** of a published empirical paper from raw WRDS data — and you **must use AI**; your use of it is itself graded. Assessed in **four moments** (weeks 39, 41, 42, 43), not one verdict at the end.

Pass: weighted average ≥ 5.5 **and** exam ≥ 5.0 . Mock exam in Week 39. Full details in the tutorial and the syllabus.

Reading, and this week

- **One puzzle-paper per week** — read it; it is exam- and discussion-relevant. Use the four-step recipe.
- **This week:** Miller (1977) (short — the clawback) and Lemmon et al. (2008) (the persistence puzzle).
- **Tutorial (Wed):** the personal-tax gain G_L as a numeric drill, and the replication-project kickoff.
- **Next week:** why conservative firms are **rational** — debt capacity as an option.

The whole course in one line: value is the benchmark, minus the frictions we will name one by one.

- Edmans, A. (2024). **“Rational Sustainability”**. In: *Journal of Applied Corporate Finance* 36.2, pp. 1–8.
- Friedman, M. (1970). ***The Social Responsibility of Business Is to Increase Its Profits***. The New York Times Magazine.
- Gelderblom, O., A. de Jong, and J. Jonker (2013). **“The Formative Years of the Modern Corporation: The Dutch East India Company VOC, 1602–1623”**. In: *The Journal of Economic History* 73.4, pp. 1050–1076.
- Graham, J. R. (2022). **“Presidential Address: Corporate Finance and Reality”**. In: *The Journal of Finance* 77.4, pp. 1975–2049.

- Hansmann, H. and R. Kraakman (2004). **“What Is Corporate Law?”** In: *The Anatomy of Corporate Law: A Comparative and Functional Approach*. Oxford University Press, pp. 1–19.
- Jensen, M. C. and W. H. Meckling (1976). **“Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure”**. In: *Journal of Financial Economics* 3.4, pp. 305–360.
- Korteweg, A. (2010). **“The Net Benefits to Leverage”**. In: *The Journal of Finance* 65.6, pp. 2137–2170.
- Le Bris, D., W. N. Goetzmann, and S. Pouget (2019). **“The Present Value Relation over Six Centuries: The Case of the Bazacle Company”**. In: *Journal of Financial Economics* 132.1, pp. 248–265.

- Lemmon, M. L., M. R. Roberts, and J. F. Zender (2008). **“Back to the Beginning: Persistence and the Cross-Section of Corporate Capital Structure”**. In: *The Journal of Finance* 63.4, pp. 1575–1608.
- Mehrotra, V. and R. Morck (2017). **“Governance and Stakeholders”**. In: *The Handbook of the Economics of Corporate Governance*. Vol. 1. Elsevier, pp. 637–683.
- Miller, M. H. (1977). **“Debt and Taxes”**. In: *The Journal of Finance* 32.2, pp. 261–275.
- (1988). **“The Modigliani–Miller Propositions After Thirty Years”**. In: *Journal of Economic Perspectives* 2.4, pp. 99–120.
- Modigliani, F. and M. H. Miller (1958). **“The Cost of Capital, Corporation Finance and the Theory of Investment”**. In: *The American Economic Review* 48.3, pp. 261–297.

Shefrin, H. (2018). ***Behavioral Corporate Finance: Decisions That Create Value***. 2nd.
McGraw-Hill Education.